

Abstract

This analysis compares the performance of four different vertical axis wind turbine designs: the Savonius, Helical 3-Blade, Double-Cup, and Quad-Cup turbines. Computational fluid dynamics (CFD) simulations were conducted using ANSYS Fluent to investigate the aerodynamic behaviour, pressure difference and maximum velocity magnitude of these turbines. The simulations were carried out at a fixed inlet velocity and temperature, which are based on the average wind speed and the average temperature in Plymouth. The results show that the Savonius, Double-Cup and Quad-Cup turbines exhibited similar values in terms of pressure difference. With Helical 3-Blade having a lower pressure difference. In terms of maximum outlet velocity, it goes from highest to lowest in the following order; Quad-Cup, Double-Cup, Helical 3-Blade and Savonius. The CFD simulation also provided insights into the flow characteristics of the turbines, with the Quad-Cup showing better flow symmetry and the least turbulence. Overall, this report highlights the importance of selecting the right vertical axis wind turbine design for specific wind conditions and emphasises the potential of CFD simulations in optimizing turbine performance.

Introduction

Wind energy is one of the fastest-growing renewable energy sources worldwide, with wind turbines being widely used for electricity generation. Vertical axis wind turbines (VAWTs) are a type of wind turbine that has a rotor shaft perpendicular to the ground. Unlike horizontal axis wind turbines (HAWTs), VAWTs are omnidirectional and can generate power from the wind coming from any direction. They are also easier to install, maintain, and operate than HAWTs. [1]

VAWTs come in various designs, each with advantages and disadvantages. The four designs that were investigated in this study are the Savonius, 3-Blade Helical, Double-Cup, and Quad Cup turbines. The Savonius turbine has a simple design and is self-starting, but it has low efficiency and generates low power [2]. The 3-Blade Helical turbine has a more complex design and higher efficiency than the Savonius, but it has a higher starting torque and is more difficult to manufacture. The Double-Cup and Quad-Cup turbines have a more complex design than the previous two but have higher efficiency and generate more power. The Quad-Cup turbine is a newer design that is gaining popularity due to its superior performance over other VAWTs. [2]

The goal of this study is to compare the performance of these four VAWT designs using CFD simulations. The simulations were performed using ANSYS Fluent software, which is a powerful tool for simulating fluid dynamics. The results were compared based on aerodynamic behaviour, pressure difference and maximum velocity magnitude.

Methodology

The analysis was performed using ANSYS Workbench and Fluent software. 3D models of the enclosure were created using DesignModeler. The enclosure was extended 1m in each direction to ensure that there is enough space for the fluid to flow. The models were then imported into ANSYS Meshing. For the meshing, the following settings were used:

- Physics Preference - CFD
- Solver Preference - Fluent
- Element Order - Program Controlled
- Element Size – Default

Below the section view of each concept design can be seen:

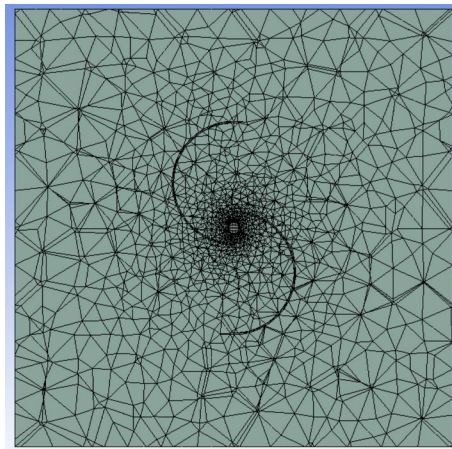


Figure x: Section view of Double-Cup mesh

Nodes - 285004

Elements - 186433

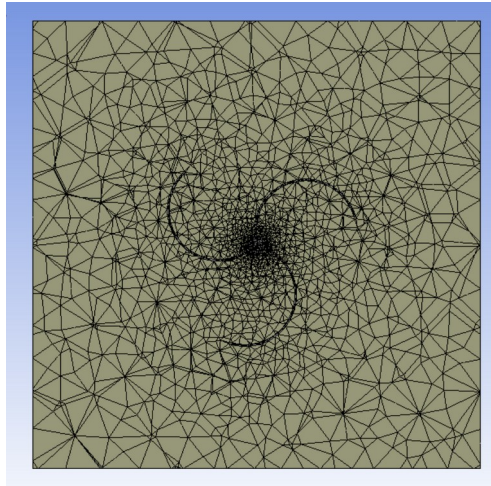


Figure x: Section view of Helical 3-Blade mesh

Nodes - 240031

Elements - 159859

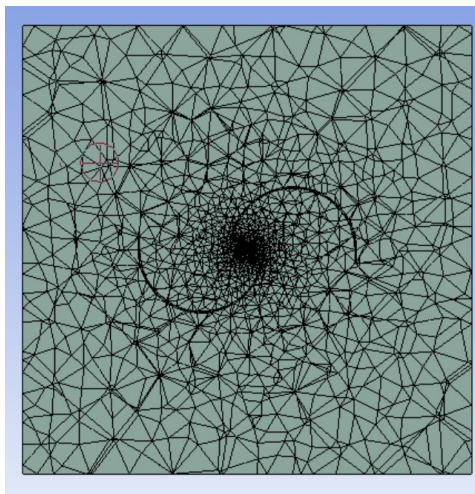


Figure x: Section view of Quad-Cup mesh

Nodes - 300675

Elements - 198561

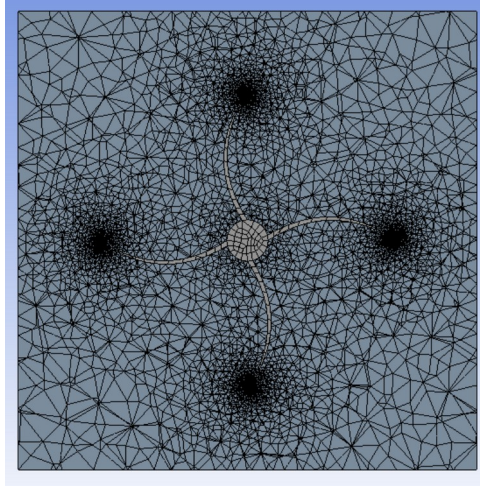


Figure x: Section view of Savonius mesh

Nodes - 1118805

Elements – 764141

The model was then imported into ANSYS Fluent for analysis. The fluid was set as air and the solid was set as aluminium. The SST k-omega turbulence model was used for the simulation. Boundary conditions were set such that the inlet velocity was 6.3m/s and the temperature was 284K at the inlet, outlet and walls of the enclosure. Hybrid initialization was used to ensure the convergence of the solution. The simulation was run for one thousand iterations. During this time, Fluent calculated the flow behaviour and temperature distribution inside the enclosure. After the simulation was complete, pressure contour and velocity magnitude path lines were created to visualize the results.

The SST k-omega turbulence model is a commonly used CFD simulation model that combines two different k-equation models to improve accuracy and robustness across a wide range of flow conditions. This model is frequently used for wind turbine analysis due to its ability to accurately simulate the complex and turbulent flow conditions surrounding the blades. The model uses the k-omega equations for the laminar inner layer of the boundary layer and the k-epsilon equations for the turbulent outer layer. Additionally, the SST k-omega model is efficient in terms of computation time, making it a suitable option for optimization studies. It is capable of simulating both attached and separated flows and is well-suited for modelling the unsteady flow conditions that are typical around wind turbines. [3]

Results

Double-Cup results:

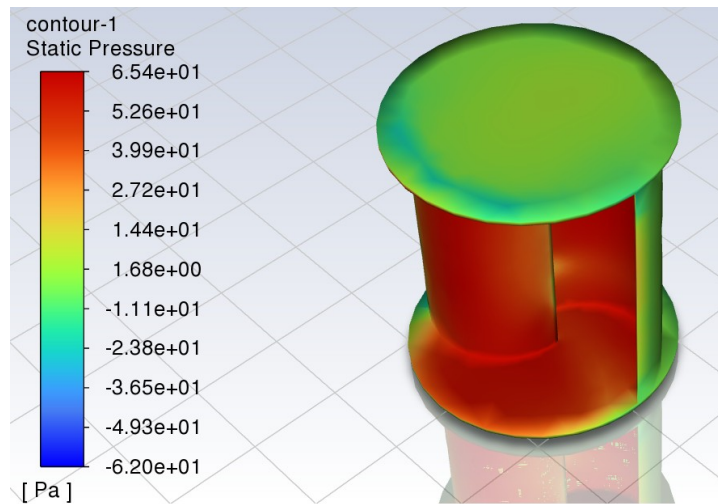


Figure x: Static Pressure contour plot of Double-Cup

Minimum static pressure = -62.02 Pa

Maximum static pressure = 65.38 Pa

Static pressure difference = 127.40 Pa

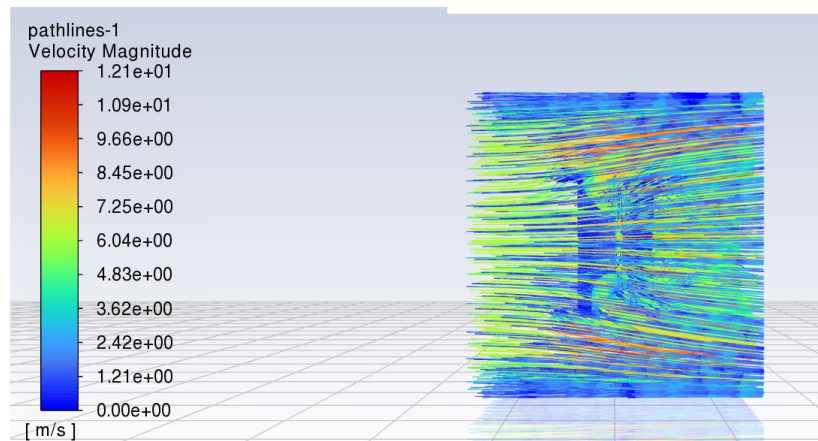


Figure x: Side view of Velocity Magnitude path lines of Double-Cup

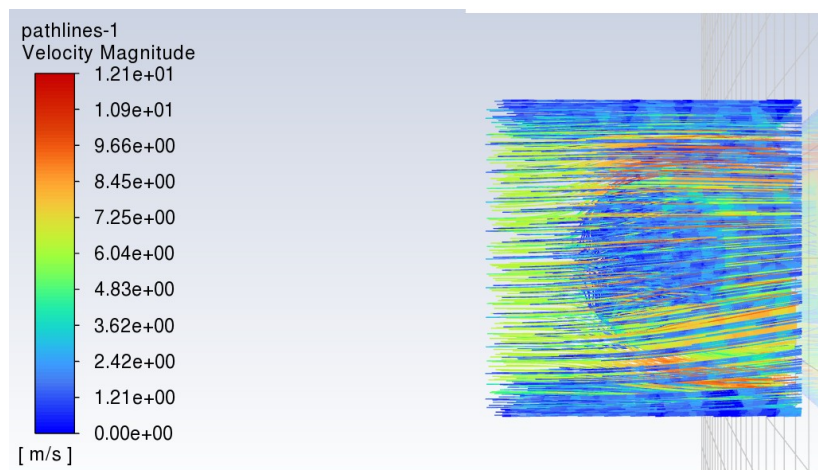


Figure x: Top view of Velocity Magnitude path lines of Double-Cup

Maximum velocity magnitude = 12.08 m/s

Drag force = 38.27 N

Helical 3-Blade results:

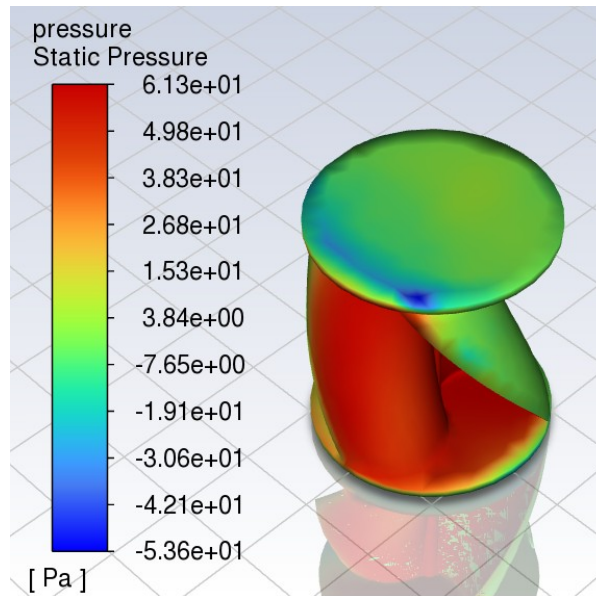


Figure x: Static Pressure contour plot of Helical 3-Blade

Minimum static pressure = -53.63 Pa

Maximum static pressure = 61.31 Pa

Static pressure difference = 114.94 Pa

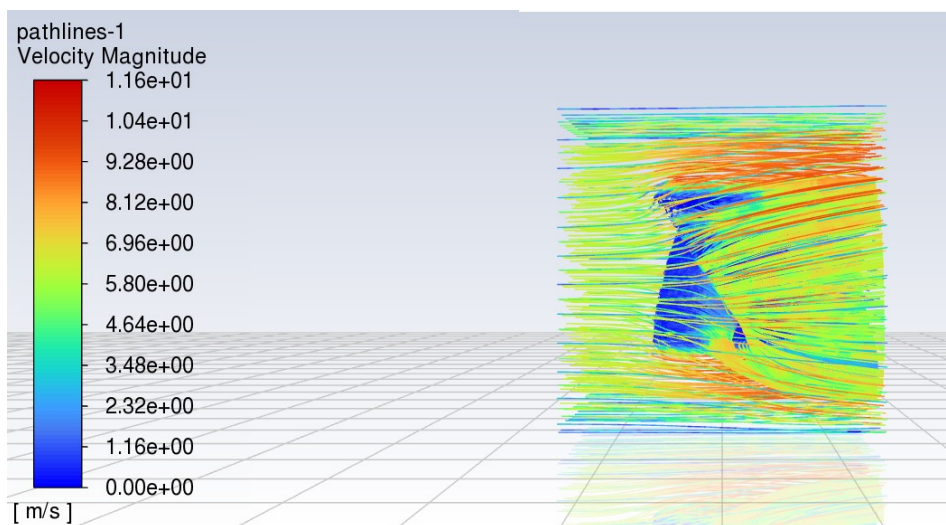


Figure x: Side view of Velocity Magnitude path lines of Helical 3-Blade

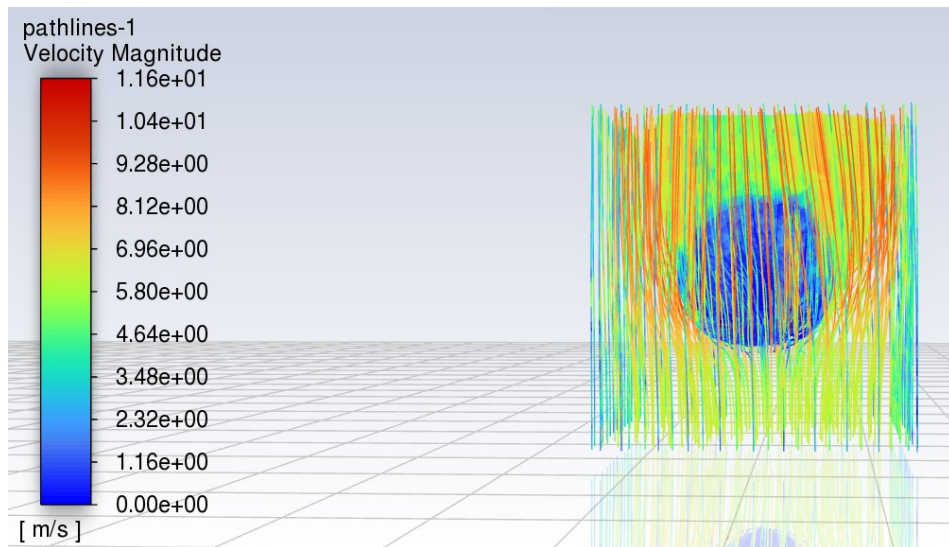


Figure x: Top view of Velocity Magnitude path lines of Helical 3-Blade

Maximum velocity magnitude = 11.60 m/s

Drag force = 38.0 N

Quad-Cup results:

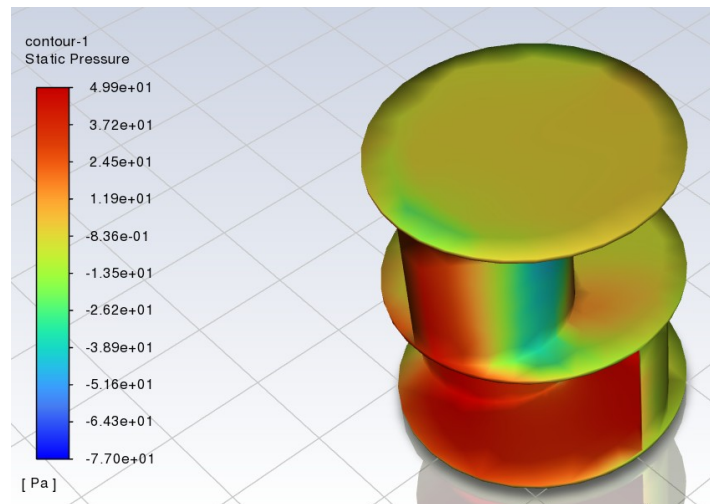


Figure x: Static Pressure contour plot of Quad-Cup

Minimum static pressure = -76.97 Pa

Maximum static pressure = 49.92 Pa

Static pressure difference = 126.89 Pa

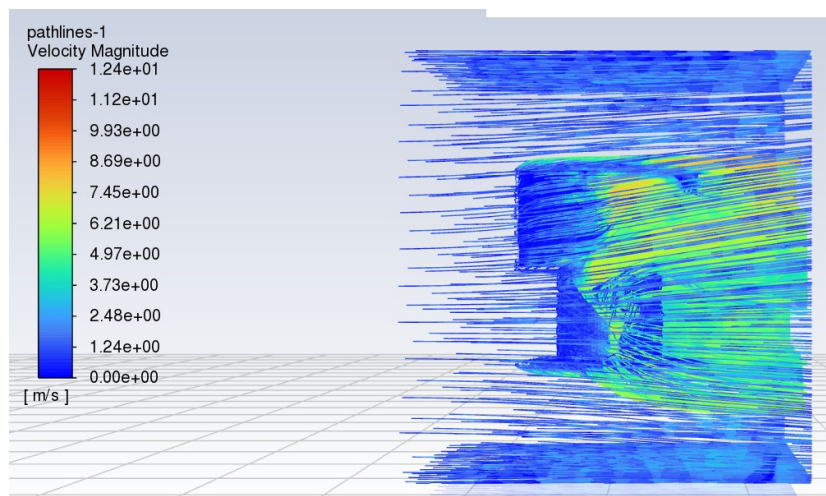


Figure x: Side view of Velocity Magnitude path lines of Quad-Cup

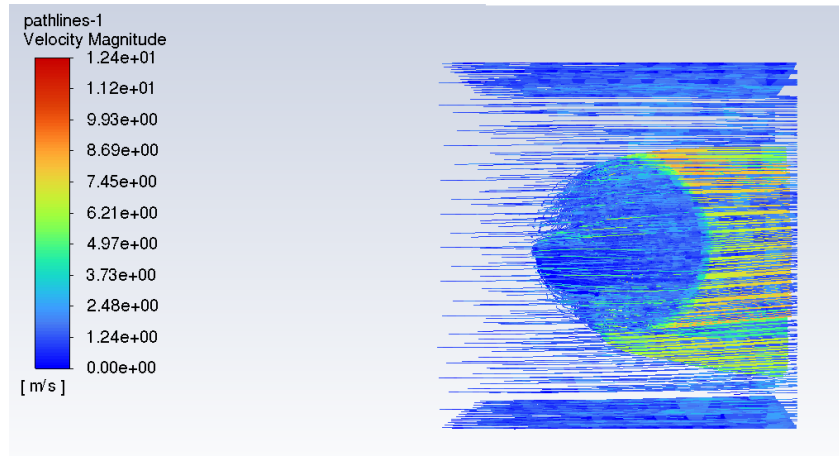


Figure x: Top view of Velocity Magnitude path lines of Quad-Cup

Maximum velocity magnitude = 12.42 m/s

Drag force = 11.13 N

Savonius results:

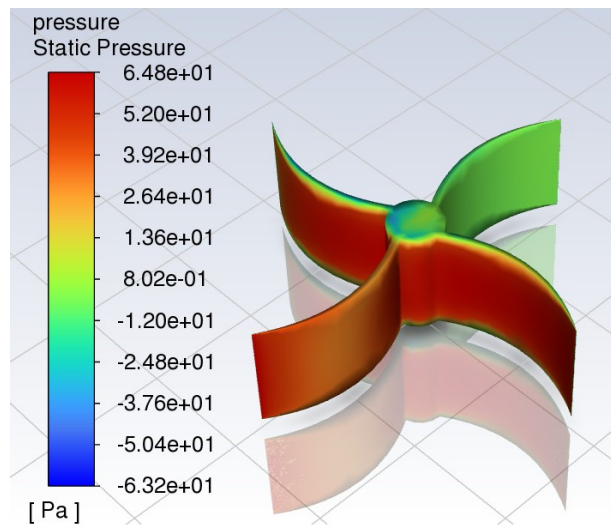


Figure x: Static Pressure contour plot of Savonius

Minimum static pressure = -63.21 Pa

Maximum static pressure = 64.82 Pa

Static pressure change = 128.03 Pa

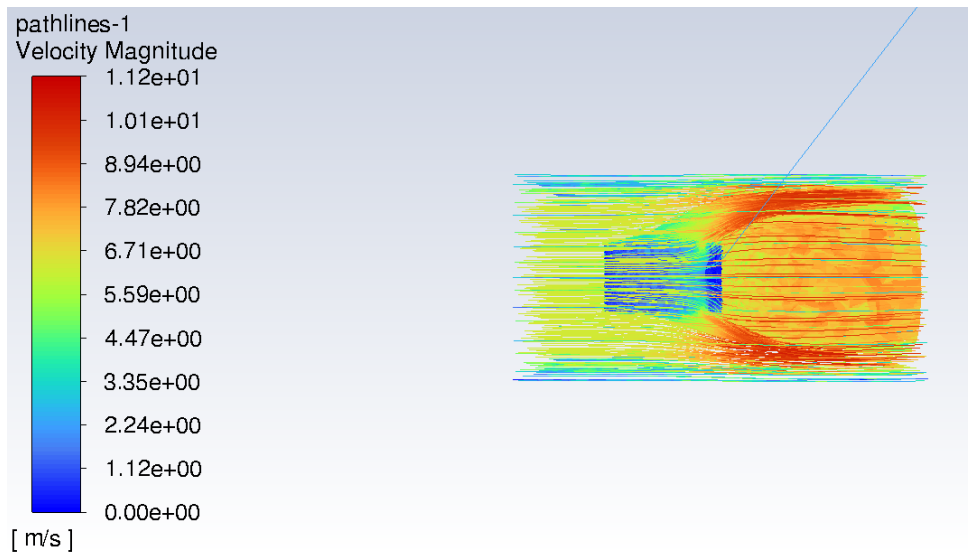


Figure x: Side view of Velocity Magnitude path lines of Savonius

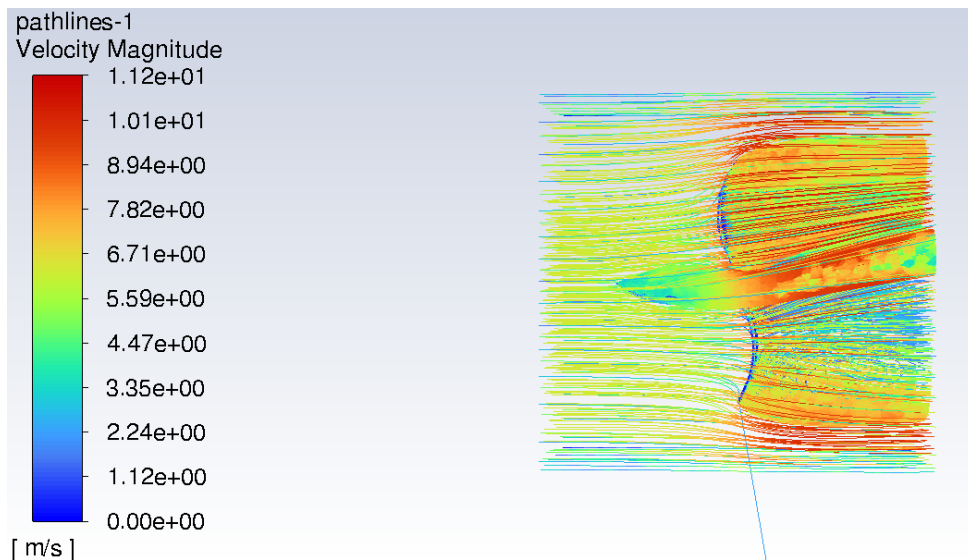


Figure x: Top view of Velocity Magnitude path lines of Savonius

Maximum velocity magnitude = 11.18 m/s

Drag force = 51.97 N

Discussion of results

After analysing the outcomes, it can be observed that the Double-Cup and Quad-Cup designs are the best choices because they possess the highest pressure difference and maximum velocity magnitude. By comparing the drag forces of both designs, it is evident that the Quad-Cup has significantly less drag force. The aerodynamic behaviour can be

analysed further by examining the velocity path lines, which indicates that the Quad-Cup is superior to the Double-Cup because of better flow symmetry and minimal turbulence. The benefits of having a less turbulent flow include:

- **Increased Efficiency:** Reducing turbulence can help decrease energy loss and enhance the overall efficiency of the turbine. Turbulent flow causes blades to experience varying and unpredictable forces that can decrease their effectiveness.
- **Improved Durability:** Turbulent flow causes wear and tear on wind turbine blades, leading to fatigue and failure. By reducing turbulence, blades undergo less stress and strain, increasing their durability and longevity.
- **Reduced Noise:** Turbulent flow can generate noise as it interacts with the blades of the turbine. Reducing turbulence can decrease the noise generated by the turbine, making it more acceptable to nearby residents.
- **Reduced Maintenance Costs:** Turbulent flow can increase maintenance costs for wind turbines as blades may require frequent repair or replacement. Reducing turbulence can decrease the amount of maintenance required, resulting in cost savings over the turbine's lifetime.

Limitations of the study:

While Computational Fluid Dynamics (CFD) has gained popularity for wind turbine analysis, there exist some constraints to CFD-based studies. These include:

- **Computational Complexity:** Wind turbines are intricate systems that involve several physical processes that are difficult to model accurately using CFD. This can necessitate high-performance computing resources and prolonged simulation times.
- **Input Accuracy:** The accuracy of CFD simulations largely depends on the precision of inputs such as turbulence intensity, wind speed, and atmospheric conditions. Incorrect inputs can lead to unreliable simulation results.
- **Numerical Method Constraints:** CFD simulations rely on numerical methods that have certain limitations, including numerical dispersion and diffusion, leading to solution errors. This is especially critical for highly transient flows, such as those present in wind turbines.
- **Validation Challenges:** The lack of experimental data makes validating CFD simulations of wind turbines difficult. This can create uncertainties about the accuracy of the simulations and limit their applicability in design optimization.
- **Simplified Geometries:** CFD studies often utilise simplified geometries, which may not precisely represent the complex geometries of actual wind turbines. This can result in inaccuracies in the simulation results.

- **Integration Challenges:** CFD is only one aspect of the wind turbine design process. To fully optimize wind turbine performance, CFD results must integrate with other design considerations such as structural analysis, electrical system design, and aerodynamic testing.

In summary, CFD-based studies can offer valuable insights into wind turbine performance and optimization, but they have limitations that require careful consideration when interpreting results.

Conclusion

In conclusion, this CFD study has provided a comprehensive comparison of the performance of four different types of vertical axis wind turbines: Savonius, 3-Blade Helical, Double-Cup, and Quad-Cup. Through a series of simulations, we have evaluated the aerodynamic characteristics of each turbine type, including the velocity magnitude and pressure difference.

Our results show that the Quad-Cup turbine has the highest maximum velocity magnitude among the four turbines, followed closely by the Double-Cup turbine. The Savonius and 3-Blade Helical turbines have lower maximum velocity magnitude, indicating lower efficiency. Additionally, the Quad-Cup turbine demonstrates relatively low drag force, which suggests lower wind resistance and increased efficiency.

These findings suggest that the Quad-Cup turbine may be the most suitable option for small-scale applications, such as residential or commercial use, where efficiency and low wind resistance are important considerations. However, the selection of a wind turbine should be based on a variety of factors, including site conditions, cost, and desired power output.

Overall, this study highlights the importance of evaluating different wind turbine types to determine the most suitable option for a given application. CFD simulations can provide valuable insights into the aerodynamic characteristics of different turbine types, allowing for informed decision-making in wind energy generation.

References

- 1 <https://www.sciencedirect.com/topics/engineering/vertical-axis-wind-turbine> M.A. Hyams, in Metropolitan Sustainability, 2012

2 <https://www.sciencedirect.com/topics/engineering/savonius-wind-turbine#:~:text=The%20major%20disadvantage%20of%20these,experimental%20rotors%20%5B11%5D>).

Ibrahim Dincer, Muhammad F. Ezzat, in Comprehensive Energy Systems, 2018

3 https://link.springer.com/chapter/10.1007/978-3-030-13543-0_3 Könözy, L. (2019). The k- ω Shear-Stress Transport (SST) Turbulence Model. In: A New Hypothesis on the Anisotropic Reynolds Stress Tensor for Turbulent Flows. Fluid Mechanics and Its Applications, vol 120. Springer, Cham. https://doi.org/10.1007/978-3-030-13543-0_3